

Algebra General Exam - August 2022

Your UVa ID Number:

- Please show all your work and justify any statements that you make.
- State clearly and fully any theorem you use.
- Vague statements and hand-waving arguments will not be appreciated.
- You may assume the statement in an earlier part proven in order to do a later part.

**DO EACH PROBLEM ON A SEPARATE SHEET OF PAPER,
AND STAPLE THEM TOGETHER IN THE CORRECT ORDER
BEFORE TURNING THE EXAM IN.**

Sign below the pledge:

"On my honor, I pledge that I have neither given nor received help on this assignment."

Problem 1

Let R denote the commutative ring $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$.

- (a) (6 points) Construct two nonassociated factorizations of 6 into products of irreducible elements (you need to verify that all elements involved in these factorizations are indeed irreducible and the two factorizations are indeed nonassociated).
- (b) (5 points) Using the factorizations from (a), construct an ideal $\mathfrak{a} \subset R$ which is not principal.
- (c) (6 points) Show that the square $\mathfrak{a}^2 = \mathfrak{a}\mathfrak{a}$ of the ideal from part (b) is principal.

Problem 2

For a complex $n \times n$ -matrix $A = (a_{ij})$ we let $\overline{A}$ denote the matrix $(\overline{a_{ij}})$ where the bar denotes the complex conjugation. Also, we let χ_A denote the characteristic polynomial of A .

- (a) (4 points) If A and $\overline{A}$ are conjugate, prove that the characteristic polynomial χ_A has real coefficients.
- (b) (7 points) Conversely, if A is diagonalizable and χ_A has real coefficients then A and $\overline{A}$ are conjugate.
- (c) (5 points) Give an example of a nondiagonalizable complex matrix A such that χ_A has real coefficients but A and $\overline{A}$ are not conjugate.

Problem 3

- (a) (7 points) Let G be a group, and $H \subset G$ be a normal subgroup. Assume that the center $Z(H)$ is $\{e\}$ and that every automorphism of H is inner. Show that G is the direct product $H \times C_G(H)$ where $C_G(H)$ is the centralizer of H in G . (Hint. Consider the action of G on H by inner automorphisms.)
- (b) (8 points) Show that there is no group G such that the commutator subgroup $[G, G]$ is isomorphic to the symmetric group S_3 . (You can assume without proof that S_3 satisfies the assumptions on H made in part (a).)

Problem 4

Let R be a commutative ring with identity. We say that a finitely generated R module P is projective if P is a direct summand of a free R -module. That is, there is some $d \geq 1$ and another R -module Q such that $P \oplus Q \simeq R^d$ as R -modules.

- (a) (6 points) Suppose P is a finitely generated projective R -module. Show that for every epimorphism of R -modules $\varphi : M \rightarrow N$, an arbitrary R -module homomorphism $\alpha : P \rightarrow N$ lifts to a homomorphism $\beta : P \rightarrow M$ such that $\varphi \circ \beta = \alpha$.
- (b) (5 points) Show that if P_1 and P_2 are finitely generated projective modules then their tensor product $P_1 \otimes_R P_2$ is also a finitely generated projective R -module.

Problem 5

- (a) (7 points) Let V and W be vector spaces over a field K , and let $v_1, \dots, v_n \in V$ be linearly independent vectors. Show that if $w_1, \dots, w_n \in W$ are such that $v_1 \otimes w_1 + \dots + v_n \otimes w_n = 0$ in $V \otimes_K W$ then $w_1 = \dots = w_n = 0$.
- (b) (7 points) Again, let V and W be vector spaces over a field K , and let $x \in V \otimes_K W$. If $x = v_1 \otimes w_1 + \dots + v_n \otimes w_n$ is a shortest presentation of x as a sum of simple tensors (i.e., x cannot be written as $v'_1 \otimes w'_1 + \dots + v'_m \otimes w'_m$ with $m < n$) then the vectors $v_1, \dots, v_n$ (and likewise $w_1, \dots, w_n$) are linearly independent.
- Note: Make sure you clearly state which properties of the tensor product you are using in both parts of this problem.

Problem 6

- (8 points) Let α be a complex number satisfying $\alpha^6 + 3 = 0$. Show that $\mathbb{Q}(\alpha) / \mathbb{Q}$ is a Galois extension, and determine its Galois group. (Hint. The same is not true if 3 is replaced by 2 .)

Problem 7

Assume that p is a prime number and $A \in GL_5(\mathbb{F}_p)$ is a matrix that satisfies $A^3 = 1$.

(a) (4 points) If $p \equiv 1 \pmod{3}$, show that A is diagonalizable.

(b) (7 points) Let $p = 11$. Classify all conjugacy classes of such matrices A .

Problem 8

(8 points) Let K be a subfield of $\mathbb{R}$, let $f(x) \in K[x]$ be an irreducible polynomial, and let L be the splitting field of f over K (i.e., the field obtained by adjoining to K all complex roots of f). Assume that the Galois group $\text{Gal}(L/K)$ is abelian. Show that if one root of f is real then all roots are real.